Chapter–2
Method

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Method

The four aims of present research "changing bases of social categorization and intergroup relationship" were to: (1) identify the dominant typologies of everyday discrimination and the role of social categorization, (2) examine how in-group and out-group identification predict social distance towards the members of out-group, (3) assess intergenerational effects on social categorization and intergroup relationship, and (4) understand the category salience and intergroup relationship in cross-categorization situations. Both qualitative and quantitative data were gathered and evaluated for the two phases: phase I, which focused on *qualitative analysis* for the accomplishment of aim 1, and phase II, which focused on *quantitative analysis* for the realization of aims 2, 3, and 4. Because phase I (qualitative analysis) builds phase II (quantitative method), all four aims in this piece of research were interrelated.

2.1 Description of Design

Research was carried out in two phases and exploratory sequential mixed method design was utilized. The mixed methods research are philosophical framework of research of inquiry that combines both qualitative and quantitative investigation models so that the findings can be combined and knowledge can be improved in a manner that is significantly higher than any one approach could accomplish on its own (Creswell & Plano Clark, 2007; Office of Behavioral and Social Sciences Research, 2001).

For first phase, to answer the questions of aim 1, exploratory phenomenological approach was employed and it is a method of inquiry generated from the schools of psychology and philosophy in which the inquirer discusses the actual experiences of people with regard to phenomena as individuals have expressed them. This description encapsulates the most important aspects of the experience for numerous people who have had similar encounters with the phenomenon. In the first phase with the help of an open-ended question, we investigated the phenomena of individuals' experiences of dominant typologies of everyday discrimination as a result of their group membership and personal characteristics. This approach has solid philosophical foundations and often involves carrying out interviews that can assist us find the various typologies across which individuals encounter discrimination on a daily basis in this modern world (Giorgi, 2009; Moustakas, 1994).

The findings from the first phase of the research were used as a foundation for the second phase of the investigation, which was dedicated to the quantitative part of the exploratory sequential strategy. The researcher did not had greater control over the manipulation of variables in the second phase of the research, and the research was conducted in a natural setting; thus, a *survey schedule* method was used in which reliable and standardized questionnaires were used to oversee the aims of the current research's second phase. We used a cross-sectional strategy to gather survey data, which implies that all of the data was gathered at one point in time.

For aim 2, we measured in-group and out-group identification of participants along with their associated segments of social distance (social distance intensity and perceived social distance) in participants' highly preferred social category. Further, we used standard multiple regression analysis to see the extent to which segments of social distance were predicted by in-group and out-group identification. Aim 3 was focused on investigating the intergenerational effects on social categorization and intergroup relationship in which researcher used generational cohorts Gen Z (born during 1997-2012), Gen Y (born during 1981-1996), Gen X (born during 1965-80), and Boomers (born during 1946-1964). For aim 4 we employed repeated measures ANOVA to understand the category salience and intergroup relationship in cross-categorization condition across whole sample and on participants' double in-group membership. Some of the study questions were better answered by utilizing qualitative approach, while others were better answered by using quantitative analysis. The pragmatic philosophy that served as the basis for this research made it possible for a methodical use of associated quantitative and qualitative techniques to address each specific aim.

2.2 Phase – I

2.2.1 Collection of Data

There were 120 adult participants with (M_{age} = 36.49 years, SD = 14.99) belonging to urban areas of Delhi (n = 60) and rural areas of Bihar (n = 60), India, participated in this study. Participants included 62 male and 58 female, and their participation was entirely voluntary; they received no monetary reward for participating in the study. Age, gender, and urban/rural distinction were taken into account as inclusion criteria; hence we used a criterion-based purposive sampling technique for selecting study participants. In qualitative research, the criteria strategy is often used to allow researchers to select an attribute or group of attributes known to vary within a population (Teddlie & Yu, 2007). Additionally, this strategy assists in collecting a more representative or comparative view of a population of interest, thereby promoting transferability, which is the ability to generalize the findings to the population under investigation (Krefting, 1991). Williams et al. (1997) developed the Everyday Discrimination Scale (EDS), which has been widely used to assess people's daily discrimination experiences. This scale consists of nine components that evaluate the subject's experiences of discrimination faced by them in their day to day life, followed by a question regarding what the subject thinks led to that daily discrimination. Our concern was not the strength of individuals' daily discriminatory experiences faced by them but rather the typologies and reasons underlying those experiences. Subjective responses of participants were written in the participants' language as expressed by participants on an open-ended follow-up question of EDS to generate qualitative data for this study. Everyday discrimination scale along with its follow-up question has been provided in Appendix–*B*.

2.2.2 Conversion of Data

Participants' written responses were transcribed into readable form, and their verbatim statements were translated into English, conveying the connotative and denotative meaning of their responses to an open-ended follow-up question of EDS. Transcription generation focused on recording persons' entire statements while excluding trivial phrases such as "well," "uh," and so on. The researcher then prepared the transcribed data for analysis in Microsoft Excel. Demographic information and each respondent's complete response to the open-ended "reasons for discrimination" EDS question were entered into columns on an Excel spread sheet, where the data was originally arranged. Each row indicated a single participant's response (N = 120).

2.2.3 Role of Researcher

The primary intent of the researcher was to gain insight into the thoughts, emotions, and personal encounters pertaining to instances of everyday discrimination encountered by the participants. This was achieved through the use of an *etic* *approach*. Getting the participants to open up about topics that can be quite personal to them was not a simple task. At times, the experiences under investigation were clearly apparent in the participant's mind, while on other occasions, remembering past experiences posed troubles. During the process of data collection, the researcher's main responsibility was to ensure the safety and confidentiality of both the participants and their data. The mechanisms implemented to ensure the protection of participants were effectively communicated and American Psychological Association Ethical Code of Conduct (APA, 2002) was followed by researcher.

2.3 Phase – II

2.3.1 Participants

In the second phase of the study, there were a total of 720 participants, ranging in age from 18 to 72 years, with a M_{age} of 37 years (SD = 14.26). Alwin (1994, 1995) addressed the rationale behind using a wider age range when he noted how each stage of the adult life cycle showed a uniformly high level of stability in his persistence model. According to this model, the projected magnitude during each of these 8-year periods shows a very high level of stability that was consistent with age on a scale from 0 to 1. The traits for which these patterns were found would appear to show a very high level of stability in individual differences throughout the course of the adult life span. There was a strong scientific basis regarding anticipating a high degree of stability after adolescence and early adulthood for many human qualities, including cognitive and intellectual capacities, certain aspects of identities, and several personality traits (Alwin, 1994, pp. 159–164). According to United Nations, Department of Economic and Social Affairs, Population Division (2015) World Population Ageing 2015 report 66.80% of participants were considered young and 33.20% were considered old. The gender distribution of participants was 51.80% male and 48.20% female. To obtain data that was truly representative of the population, we conducted a survey with native participants whose mother tongue was the language spoken in their area. We sampled 120 participants from each of the following states and one union territory: Assam (North-East Zone), Bihar (Eastern Zone), Gujarat (Western Zone), Telangana (South-Central Zone), Uttar Pradesh (North Central Zone), and Delhi (North Zone). Our intention was not to compare the individuals from the various locations on different measures but rather to get a better representation of data in the Indian context. These zones were selected based on the

Pew Research Centre's (2021) report "Religion in India: Tolerance and Segregation," which defined regions of India in different zones using the Census of India, 2011 (p. 15); states and union territory were identified based on NCRB findings from "Crime in India" (NCRB, 2021, August 29).

With a medium effect size (ω = .30, which corresponds to the 50th percentile of the effect size distribution), a *G* Power analysis* revealed that 243 participants were enough to have a 95% chance of obtaining the same or larger effect size. To determine the sample size for an extensive population survey in the social sciences, we used the calculations from Fowler's (2009) table, considering a margin of error of ± 5%, a confidence interval of 95%, and a probability of 50% that characteristics of our target population will be included in the sample. After taking into account the statistical output from G* Power and the recommendation made by Fowler, we concluded that a sample size of 720 would provide us with an adequate response rate and a better representation of the population's characteristics. The stratified random sampling approach was used for the second phase of sampling. To reduce the error of respondents' literacy and language limitations, the data was collected face to face utilizing a survey schedule approach. Table 2.1 displays the sample's characteristics.

Inclusion criteria

- Adult participants equal and above the age of 18 years.
- Native people who had casted their vote in that state or UT at least twice and knew their mother tongue as their primary language.
- Participants who gave consent to participate in the research.

Exclusion criteria

- Participants below the age of 18 years.
- Those people who were non-native and recently shifted to that state or UT.
- Members of LGBTQ+ communities excluded because of dichotomous nature of cross-categorization condition.
- Members who refused to provide consent were excluded.

Table 2.1

Sample Characteristics

Socio-demographics	*f*	%
Age		
Young (18-44 years)	481	66.80
Old (above 44 years)	239	33.20
Gender		
Male	373	51.80
Female	347	48.20
Residential status		
Urban	432	60.00
Rural	288	40.00
Religious orientation		
Religious	606	84.20
Non-religious	114	15.80
Caste		
Unreserved	403	56.00
Reserved	317	44.00
Educational level		
Illiterate	268	37.20
Literate	452	62.80
Economic status		
Economically strong	333	46.30
Economically weak	387	53.80

Note. N = 720.

Table 2.1 reflected that in the sample, there were people who reported residing in urban and rural areas, with 60% of participants stating that they lived in urban areas and 40% stating that they lived in rural areas. Additionally, the participants belonged to different generational cohorts. The majority of people (84.20%) claimed to be religious, while 15.80% reported being non-religious. 56% of the overall sample claimed that they belonged to the unreserved category, while the remaining 44% were members of the reserved category. When asked about their educational level, 37.20% of individuals said they were illiterate, while 62.80% said they were literate. Furthermore, 46.30% of participants reported having strong economic living conditions, whereas 53.80% reported being economically weak.

2.3.2 Measures

2.3.2.1 Socio-Demographics. Age, gender, State/UT of belongingness, residential status (urban/rural), religion, and religious orientation (religious or non-religious), caste (unreserved or reserved), native language, preferred language, educational level (illiterate or literate) and economic status (economically strong or economically weak) of the participants were collected. This information was considered ideal in order to identify and indicate participants' in-group and out-group in their highly preferred social category and also they could understand the scenarios in which they shared an in-group or out-group membership to a social group on various social categories. Appendix–*C* contains socio-demographic sheet. The detailed information regarding participants' in-group and out-group based on his/her preferred social category has been provided in Appendix–*H*.

2.3.2.2 Social Categories Preference Question (SCPQ). Based on findings of phase I, we selected eight highly occurring typologies of discrimination as social categories to work with in phase II, based on Jaccard's coefficient lifestyle and race and ethnicity were excluded, also physical appearance was not considered as an indicative of social category. We investigated the preferences of participants for their preferred social category in order of preference in their life. We arranged these social categories in unstructured manner within rows and participants were asked to rank from 1 to 8, where rank 1 was the most preferred while 8 was least preferred. Hence, participants were required to provide rank by priority from 1-8. Krippendorff α inter-rater reliability for this measure was found to be .88 ($\alpha > .80$) on whole sample of this study suggesting strong inter-rater reliability in terms of ranking of social categories and their preferences. This measure was also used as manipulation check. Social categories preference question has been placed as Appendix–*D*.

2.3.2.3 Group Identification Scale (GIS). Doosje et al. (1995) group identification scale consisting 4 items was used to measure in-group and out-group group identification of participants with respect to their highly preferred social category for aim 2 and 3. For instance, if a literate person selects education as his or her preferred social category, that person's in-group will be members of the literate group, and his or her out-group will be members of the illiterate group. However, if an illiterate person selects education as his or her preferred social category, that person's in-group will be members of the illiterate group, and his or her out-group will be members of the literate group. The items of GIS were 'I identify with other [members of the group]', 'I see myself as a member of [the group]', 'I am glad to be a member of [the group]', and 'I feel strong ties with other [members of the group]' in which the items' framing can be changed by substituting [the group] with the specific group (e.g., my team, my organization, explicit name of group). On the scale that was employed, responses ranged from 1 (strongly disagree) to 7 (strongly agree), and the Likert scale had seven points in total. We used GIS twice, once for measuring in-group identification and another for measuring out-group identification by substituting [the group] with respective in-group and out-group of participants' highly preferred social category. The four items group identification scale formed a reliable scale of in-group and out-group identification. In this research the Cronbach's α for in-group identification scale on whole sample was found to be .75 and for out-group identification it was .88 suggested fairly good reliability. Area wise Cronbach's α for in-group identification was found to be in the range of .71 to .81., whereas for out-group identification Cronbach's α was found to be in the range of .83 to .90 which revealed that both the scale provided consistent results and worked considerably well in each area. Participants' preferred social category wise Cronbach's α for in-group identification was found to be in the range of .72 to .87 and for out-group identification it was found to be in the range of .76 to .88 suggesting both the scales provided reliable measures to assess in-group and out-group identification of participants on their preferred social category. In-group and out-group identification scales have been given in Appendix–*E*.

2.3.2.4 Social Distance Scale (SDC). Modified social distance scale was used in aim 2 and 3 which was developed by Mather et al. (2017). This scale utilizes original Bogardus Social Distance Scale (1925a) that measures the perceived social distance between people on many factors likes religion, ethnicity, gender etc. and used to study prejudice and intergroup relations through employing Guttman's rank order scaling system. In this adapted scale, the authors also introduced a Social Distance Intensity Score (iScore), which served as an intensified metric that uniquely combined the Bogardus and Likert scales through multiplication. The scoring process involved reversing the coding for items 1 to 6 and then applying the rank order from the Bogardus scale as a multiplier to the Likert responses for each item. An illustration of this can be found in the Bogardus statement, which expresses the individual's willingness to accept a specific group member as a close relative through marriage. This particular statement signifies the minimal degree of social distance and is consequently assigned a ranking of 1 on the original Bogardus scale. The rank one value was subsequently multiplied by the Likert score provided by the participant.

Therefore, a participant who expresses a strong agreement in accepting individuals from a particular group as potential relatives through marriage would be assigned a score of 1 (1×1) for that specific item. The act of excluding a member of a group from the country signifies the manifestation of the utmost degree of social distance, as it was assigned a ranking of 7. Consequently, an individual who exhibited a solid inclination to exclude a particular individual from a specific group would be given a social distance score of 35 (7 multiplied by 5) for that particular item. The sums of the responses for each item were taken together to calculate the overall iScore for a particular category. The total iScore exhibited a range of 28 to 140. Thus this modified scale provided two segments of social distance first social distance intensity score and second perceived social distance score between individuals of different groups. This scale was used only to measure the social distance towards the '*out-group members'* on individuals' highly preferred social category. In this research the Cronbach's α for social distance intensity measure on whole sample was found to be .86. Area wise Cronbach's α for social distance intensity measure was found to be in the range of .84 to .88. Participants' preferred social category wise Cronbach's α for social distance intensity measure was found to be in the range of .74 to .88. Krippendorff α inter-rater reliability for perceived social distance measure on whole sample was found to be .86 ($\alpha > .80$) suggesting strong inter-rater reliability in terms of perception of social distance towards member of out-group. The social distance scale has been provided in Appendix–*F*.

2.3.2.5 Social Identification Scale (SIS). In order to examine the salience of categories in Aim 4, the researcher employed a social identification measure known as the Single-Item Social Identification (SISI) statement, as provided by Postmes et al. (2013). This measure was utilized to evaluate the overall social identification of participants towards various social groups that emerged in the context of *crossed-categorization condition (CCC)*. Additionally, the researchers assessed participants' social identification by considering their double in-group membership. The measure included response to one's agreement with the phrase "I identify with members of the group (or category)", on a 7-point scale, 1 for not at all and 7 for completely. Three investigations by Postmes et al. (2013) found that the SISI was valid and reliable across a wide variety of social groups. The Social Identification Scale, along with various cross-categorization conditions, has been incorporated into Appendix–*G*.

2.3.3 Procedure

Permission for data collection was taken from the Research Advisory Committee (RAC) and Board of Studies (BoS), department of Psychology, Aligarh Muslim University, Aligarh, (India). Researcher located and resided at the places from where he had to collect the data from 1st August 2022 to 30th November 2022. The researcher visited hospitals, parks, government offices, etc., and asked people from different strata to participate in the study after establishing a rapport with the participants and providing them with background information about the study's purpose. Data was collected through survey schedule method and the participants were requested to sit before the researcher. The survey schedule commenced by administering an informed consent form to participants, who were requested to provide their signature as an indication of consent (the consent form can be found in *Appendix–A* of this research work). The consent form explicitly stated that the confidentiality of participants' responses would be maintained and their participation would be entirely voluntary. Access to the main section of the survey schedule was granted exclusively to participants who provided their consent to participate. Instructions were given to the participants and schedule started with socio-demographic variables, followed by the list of preference of eight categorization dimensions formed based on findings of phase one, and measures of importance to research used in second phase. On an average survey schedule took 12-15 minutes. Participants were acknowledged for their time and effort at the end of the survey schedule, and any questions or concerns they may have had about the study were addressed.

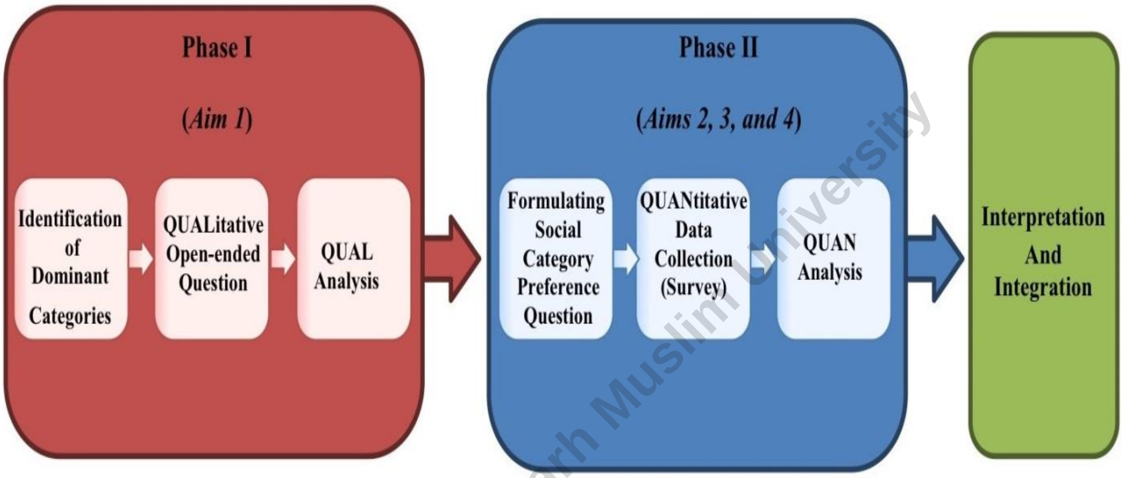
2.3.4 Data Analysis Strategy

All the data collected in second phase of research were entered into Microsoft Excel. IBM-SPSS (Statistical Package for Social Sciences) software version 28 (IBM, 2021) was used for the analysis. Before proceeding to the final analysis, missing values, outliers, range, and inflated standard deviations were examined. Appropriate statistical tests were employed to analyse the descriptive and inferential statistics, as well as their effect size, that were required to fulfil the aims and their associated objectives of the second phase of research. The results were given in the result section of this thesis in accordance with the requirements of the 7th edition of the APA publication manual (APA, 2020) and the findings were discussed in the discussion section of this thesis in accordance with the existing literature and theories.

The schema to analyse the data prepared and followed at various stages of research has been displayed by figure 2.1.

Figure 2.1

Schema of Analysis in Two Phases (Implementation Matrix)



2.4 Ethical Considerations

The research proposal was submitted to and accepted by the Board of Studies (BoS) and Committee for Advanced Studies and Research (CASR) of Aligarh Muslim University. American Psychological Association Ethical Code of Conduct (APA, 2002) guidelines were followed at each step of research, data collection, analysis and reporting of results. In addition, prior to approaching the respondents, approval to collect data was first acquired from the Chairperson of the Department of Psychology who put forward data collection request in BoS. Before taking part in the research, the participants were made aware of the purpose of the research and asked to give their consent to participate also there was no intent to harm any individual during the research. The participants were assured of their confidentiality and free will. To provide the research community with a better understanding and greater clarity, the results were reported using the 7th edition of the APA publication manual (APA, 2020).